

AWS S3 Complete Study Guide: Concepts, Features, Examples & Interview Questions

Amazon S3 (Simple Storage Service)

Definition

Amazon S3 (Simple Storage Service) is a highly scalable, durable, secure, and cost-effective **Object Storage Service** provided by AWS. It is used to store and retrieve any amount of data from anywhere on the internet. S3 stores data as **Objects** inside **Buckets**. It provides **99.999999999% (11 Nines) Durability**, supports unlimited storage, integrates with almost all AWS services, and is widely used for backups, data lakes, static website hosting, log storage, disaster recovery, and application data storage.

Key Points of S3

1. Object Storage Service
 2. Stores data as Objects
 3. Objects are stored inside Buckets
 4. Bucket names must be globally unique
 5. Unlimited Storage Capacity
 6. 11 Nines Durability
 7. Highly Scalable
 8. Highly Available
 9. Secure with IAM and Bucket Policies
 10. Supports Encryption
 11. Supports Versioning
 12. Supports Lifecycle Policies
 13. Supports Replication
 14. Supports Event Notifications
 15. Can Host Static Websites
 16. Supports Multipart Upload
 17. Integrated with EC2, Lambda, Athena, EMR, Redshift, etc.
-

Types of S3 Storage Classes

AWS S3 has multiple storage classes optimized for different access patterns and costs.

1. S3 Standard

Definition

S3 Standard is the default storage class designed for frequently accessed data. It provides low latency, high throughput, and high availability. It is suitable for applications where data is accessed regularly.

Example

- Websites
- Mobile Apps
- Frequently Used Files
- Application Data

Features

- Multi-AZ Storage
 - Low Latency
 - High Performance
 - High Availability
-

2. S3 Standard-Infrequent Access (Standard-IA)

Definition

S3 Standard-IA is designed for data that is accessed less frequently but requires immediate access when needed. It provides lower storage costs than Standard but includes retrieval charges.

Example

- Monthly Reports
- Backup Files
- Disaster Recovery Data

Features

- Lower Cost
 - Millisecond Access
 - Multi-AZ Storage
 - Retrieval Fee Applies
-

3. S3 One Zone-IA

Definition

S3 One Zone-IA stores data in only one Availability Zone instead of multiple AZs. It is cheaper than Standard-IA but has less redundancy.

Example

- Secondary Backups
- Re-creatable Data
- Temporary Archives

Features

- Single AZ Storage
 - Lower Cost
 - Fast Retrieval
 - Less Fault Tolerance
-

4. S3 Intelligent-Tiering

Definition

S3 Intelligent-Tiering automatically moves objects between storage tiers based on access patterns. It helps reduce costs when access frequency is unpredictable.

Example

- User Uploaded Files
- Unknown Access Patterns
- Long-Term Storage

Features

- Automatic Tier Movement
 - Cost Optimization
 - No Performance Impact
-

5. S3 Glacier Instant Retrieval

Definition

Designed for archive data that is rarely accessed but requires instant retrieval when needed.

Example

- Medical Records
- Archived Images

- Historical Documents

Features

- Archive Storage
 - Millisecond Retrieval
 - Lower Cost than Standard
-

6. S3 Glacier Flexible Retrieval

Definition

Low-cost archival storage where retrieval times range from minutes to hours.

Example

- Old Backups
- Archived Reports
- Compliance Data

Features

- Very Low Cost
 - Retrieval in Minutes or Hours
 - Long-Term Storage
-

7. S3 Glacier Deep Archive

Definition

The lowest-cost storage class in AWS designed for long-term archival data that is rarely accessed.

Example

- Tax Records
- Legal Documents
- 7–10 Year Archives

Features

- Lowest Storage Cost
 - Retrieval up to 12 Hours
 - Long-Term Retention
-

Storage Class Comparison

Storage Class	Access Frequency	Retrieval Speed
Standard	Frequent	Milliseconds
Standard-IA	Infrequent	Milliseconds
One Zone-IA	Infrequent	Milliseconds
Intelligent-Tiering	Variable	Milliseconds
Glacier Instant Retrieval	Rare	Milliseconds
Glacier Flexible Retrieval	Rare	Minutes–Hours
Glacier Deep Archive	Very Rare	Hours

Important Interview Points

What is S3?

Amazon S3 is a highly scalable object storage service used to store and retrieve data from anywhere.

What is a Bucket?

A Bucket is a container that stores objects.

What is an Object?

An Object is a file stored inside an S3 bucket.

What is a Key?

A Key is the unique name/path of an object.

What is the Maximum Object Size?

5 TB

What is the Direct Upload Limit?

5 GB

What is Versioning?

Maintains multiple versions of an object.

What is Lifecycle Policy?

Automatically moves or deletes objects based on rules.

What is Replication?

Automatically copies objects to another bucket or region.

What is Encryption?

Protects data stored in S3.

Quick Revision

Amazon S3 = Object Storage

Storage Classes:

1. Standard
2. Standard-IA
3. One Zone-IA
4. Intelligent-Tiering
5. Glacier Instant Retrieval
6. Glacier Flexible Retrieval
7. Glacier Deep Archive

Maximum Object Size = 5 TB

Direct Upload = 5 GB

Bucket Storage = Unlimited

Durability = 99.999999999% (11 Nines)

AWS S3 Versioning - Complete Notes

What is S3 Versioning?

Definition

S3 Versioning is a feature that allows Amazon S3 to keep **multiple versions of the same object** in a bucket. Whenever a file is modified, overwritten, or deleted, S3 preserves the previous version instead of permanently removing it.

Example

Without Versioning:

resume.pdf

↓ overwrite

resume.pdf (old file lost)

With Versioning:

resume.pdf (v1)

resume.pdf (v2)

resume.pdf (v3)

Old versions remain available.

Why Use Versioning?

Definition

Versioning protects data from:

- Accidental deletion
- Accidental overwrite
- Application failures
- User mistakes

Example

Employee uploads wrong file:

report.xlsx (v1)

report.xlsx (v2 - wrong file)

You can restore v1.

Versioning States

1. Enabled

Definition

Versioning is active and every object modification creates a new version.

Example

photo.jpg (v1)

photo.jpg (v2)

photo.jpg (v3)

2. Suspended

Definition

Stops creation of new versions but keeps all old versions.

Example

Existing:

v1

v2

v3

Versioning Suspended

Old versions still exist.

Can Versioning Be Disabled?

Definition

No. Once enabled, Versioning cannot be permanently disabled.

Only:

Suspend Versioning

is allowed.

Version ID

Definition

Every object version receives a unique Version ID.

Example

resume.pdf

Version ID:

123abc

456def

789ghi

Each version has a unique ID.

Delete Marker

Definition

When you delete an object in a versioned bucket, S3 creates a Delete Marker instead of deleting all versions.

Example

resume.pdf (v1)

resume.pdf (v2)

Delete Object

Result:

Delete Marker

v1

v2

Object appears deleted but old versions still exist.

Recover Deleted Objects

Definition

Deleted objects can be recovered by removing the Delete Marker.

Example

Delete Marker Removed

Object becomes visible again.

Permanent Deletion

Definition

To permanently delete data, specific versions must be deleted.

Example

Delete v1

Delete v2

Delete v3

Then object is permanently removed.

Versioning and Storage Cost

Definition

Every version consumes storage space and incurs charges.

Example

File Size = 10 MB

v1 = 10 MB

v2 = 10 MB

v3 = 10 MB

Total Storage:

30 MB

AWS charges for all versions.

Versioning with Lifecycle Rules

Definition

Lifecycle Policies can automatically delete old versions.

Example

Keep Versions for 90 Days

After 90 Days

Delete Old Versions

MFA Delete

Definition

Provides extra security by requiring MFA (Multi-Factor Authentication) before deleting versions.

Example

Delete Object

Requires:

Password + OTP

Versioning and Replication

Definition

When Replication is enabled, object versions can be copied to another bucket.

Example

Mumbai Bucket



Singapore Bucket

All versions replicated.

Versioning and Encryption

Definition

Each version can be encrypted independently.

Example

v1 → Encrypted

v2 → Encrypted

v3 → Encrypted

Versioning Use Cases

Backup

Protect important files.

Example

Database Backups

Disaster Recovery

Recover deleted files.

Example

Accidental Deletion

Compliance

Maintain historical records.

Example

Financial Documents

Application Data Protection

Protect application files.

Example

Configuration Files

How to Enable Versioning?

AWS Console

S3



Bucket



Properties



Bucket Versioning



Enable

Interview Questions

What is S3 Versioning?

A feature that keeps multiple versions of an object.

Why use Versioning?

To protect against accidental deletion and overwrite.

Can Versioning be disabled?

No, it can only be suspended.

What is a Version ID?

Unique identifier assigned to each version of an object.

What is a Delete Marker?

A marker created when an object is deleted in a versioned bucket.

Does Versioning increase cost?

Yes, because every version consumes storage.

Can deleted files be recovered?

Yes, by restoring previous versions or removing the Delete Marker.

Real-World Example

Suppose you have:

employee_data.xlsx

Day 1:

Version 1

Day 2:

Version 2

Day 3:

Version 3

Day 4:

File Deleted

S3 creates:

Delete Marker

If needed:

Remove Delete Marker

File is restored instantly.

Quick Revision

Versioning = Multiple Versions of Objects

Benefits:

- ✓ Recover Deleted Files
- ✓ Recover Overwritten Files
- ✓ Data Protection
- ✓ Backup Support
- ✓ Disaster Recovery

States:

1. Enabled
2. Suspended

Cannot:

- ✗ Permanently Disable

Important Terms:

- Version ID
- Delete Marker
- MFA Delete
- Lifecycle Rule
- Replication

. What is S3 Versioning?

Answer

S3 Versioning is a feature that keeps multiple versions of the same object in a bucket. It helps protect data from accidental deletion or overwriting. Every time an object is updated, S3 creates a new version instead of replacing the old one. This improves data recovery and backup capabilities. It is commonly used for important business files.

Example

report.xlsx (v1)
report.xlsx (v2)
report.xlsx (v3)

2. Why is Versioning used in S3?

Answer

Versioning is used to protect data from accidental deletion and modification. It allows users to recover previous versions of files whenever needed. Organizations use it for backups, compliance, and disaster recovery. It ensures that important data is not permanently lost. Versioning improves data reliability.

Example

An employee accidentally uploads the wrong Excel file. The previous version can be restored.

3. How do you enable Versioning?

Answer

Versioning can be enabled through the AWS Management Console, AWS CLI, or SDKs. Once enabled, S3 starts assigning unique Version IDs to objects. All future updates create new versions automatically. Existing objects also receive version IDs. It is configured at the bucket level.

Example

S3 → Bucket → Properties → Versioning → Enable

4. Can Versioning be disabled?

Answer

No, Versioning cannot be permanently disabled after it has been enabled. AWS only allows you to suspend Versioning. When suspended, existing versions remain available. New uploads will not create new versions in the same way. This design prevents accidental loss of historical data.

Example

Enabled → Suspended

5. What is a Version ID?

Answer

A Version ID is a unique identifier assigned to every version of an object. AWS uses this ID to distinguish different versions of the same file. It helps retrieve specific versions when needed. Each version receives a unique ID automatically. Users can use the Version ID to restore data.

Example

Version 1 = abc123

Version 2 = xyz456

6. What happens when you overwrite an object in a versioned bucket?

Answer

When an object is overwritten, S3 does not replace the old file. Instead, it creates a new version and keeps the previous version intact. This allows users to access both old and new copies. It prevents accidental data loss. Recovery becomes very easy.

Example

file.txt (v1)

file.txt (v2)

7. What happens when you delete an object in a versioned bucket?

Answer

When an object is deleted, S3 creates a Delete Marker instead of removing all versions. The object appears deleted to users. However, older versions still exist in the bucket. These versions can be restored later if required. This provides protection against accidental deletions.

Example

Delete Marker

v1

v2

8. What is a Delete Marker?

Answer

A Delete Marker is a special object created when an object is deleted from a versioned bucket. It becomes the current version and hides previous versions. The actual data is not removed. Removing the Delete Marker restores visibility of older versions. It acts like a soft delete.

Example

report.pdf

↓ Delete

Delete Marker Created

9. How can you recover a deleted object?

Answer

A deleted object can be recovered by deleting the Delete Marker or restoring a previous version. Since older versions remain stored, recovery is simple. Administrators can access the version history. This feature is useful during accidental deletions. Recovery is almost immediate.

Example

Delete Marker Removed

→ File Restored

10. Does Versioning increase storage cost?

Answer

Yes, Versioning increases storage costs because AWS charges for every version stored. Each version consumes storage space. If a file is updated frequently, costs can grow significantly. Lifecycle policies are often used to manage old versions. Proper monitoring is recommended.

Example

10 MB File

v1 = 10 MB

v2 = 10 MB

v3 = 10 MB

Total = 30 MB

11. What is MFA Delete?

Answer

MFA Delete adds an extra security layer to S3 Versioning. Users must provide Multi-Factor Authentication before deleting object versions. This prevents unauthorized deletions. It is especially useful for sensitive data. It improves security and compliance.

Example

Password + OTP

Required for Delete

12. What is the relationship between Versioning and Lifecycle Policies?

Answer

Lifecycle Policies can automatically manage old versions of objects. They can move versions to Glacier or delete them after a specific period. This helps reduce storage costs. It is commonly used in backup solutions. Lifecycle rules work well with Versioning.

Example

90 Days → Glacier

365 Days → Delete

13. Does Versioning work with Replication?

Answer

Yes, Versioning is required for Cross-Region Replication (CRR) and Same-Region Replication (SRR). AWS replicates object versions between buckets. This improves disaster recovery and data availability. Replication ensures copies exist in other locations. It is widely used by enterprises.

Example

Mumbai Bucket



Singapore Bucket

14. Can you restore an old version?

Answer

Yes, any previous version can be restored by making it the current version. Users can download, copy, or promote older versions. This is useful when incorrect changes are made. Restoration is simple and quick. Historical versions remain accessible.

Example

v1 → Restore

Current Version

15. What is the versioning state of a new bucket?

Answer

A newly created bucket has Versioning disabled by default. AWS does not automatically enable it. Users must explicitly enable Versioning if required. This allows organizations to choose their storage strategy. Versioning is optional.

Example

New Bucket

Versioning = Off

16. Is Versioning enabled per object or per bucket?

Answer

Versioning is enabled at the bucket level, not at the object level. Once enabled, all objects inside the bucket are affected. Every upload receives a version ID. There is no option to enable Versioning for selected files only. Bucket-wide configuration simplifies management.

Example

Bucket A

→ All Objects Versioned

17. What are common use cases of Versioning?

Answer

Versioning is used for backups, disaster recovery, compliance requirements, document management, and application data protection. It provides recovery from accidental changes. Many organizations

use it for critical business files. It supports auditing requirements. It is a best practice for important data.

Example

HR Records
Financial Documents
Database Backups

18. What happens if Versioning is suspended?

Answer

When Versioning is suspended, old versions remain stored. New uploads do not create fully versioned copies in the same way. Existing historical versions are preserved. Users can still access and restore older versions. Suspension does not delete existing versions.

Example

v1
v2
v3
Versioning Suspended

19. How does Versioning help in Disaster Recovery?

Answer

Versioning protects data from accidental deletion, corruption, and overwrites. Organizations can restore previous versions quickly. Combined with Replication, it provides a strong disaster recovery strategy. Data remains available even after mistakes. Recovery time is significantly reduced.

Example

Deleted Backup
↓
Restore Previous Version

20. What are the advantages of S3 Versioning?

Answer

Versioning provides data protection, easy recovery, audit capabilities, backup support, and compliance benefits. It prevents permanent data loss from human errors. Historical versions remain accessible. It integrates with Lifecycle Policies and Replication. It is considered a best practice for important S3 buckets.

Example

Benefits:
✓ Backup

- ✓ Recovery
- ✓ Compliance
- ✓ Audit Trail
- ✓ Disaster Recovery

AWS S3 MFA Delete - Interview Questions and Answers

1. What is MFA Delete in Amazon S3?

Answer

MFA Delete is an additional security feature of S3 Versioning that requires Multi-Factor Authentication (MFA) before permanently deleting an object version or changing the versioning state of a bucket. It helps protect critical data from accidental or unauthorized deletion. Even if someone has AWS credentials, they cannot delete protected versions without the MFA device. MFA Delete is mainly used for sensitive and compliance-related data.

Example

User tries to delete file

↓

AWS asks for OTP

↓

Delete allowed

2. Why is MFA Delete used?

Answer

MFA Delete is used to add an extra layer of security to versioned buckets. It prevents hackers, administrators, or users from accidentally deleting important data. Organizations use it to protect backups, financial records, and compliance documents. It reduces the risk of ransomware attacks and human errors. MFA Delete ensures that a physical MFA device is required before destructive actions.

Example

Backup Files

↓

Protected by MFA Delete

3. Which operations require MFA Delete?

Answer

MFA Delete is required for two important operations:

1. Permanently deleting an object version.
2. Changing the versioning state of a bucket.

Normal object uploads and downloads do not require MFA Delete. Only sensitive actions need MFA verification.

Example

Delete Version v2

↓

OTP Required

4. Can MFA Delete be enabled without Versioning?

Answer

No. MFA Delete requires Versioning to be enabled first. Since MFA Delete protects object versions, it cannot work on non-versioned buckets. AWS enforces this dependency. If Versioning is not enabled, MFA Delete cannot be configured.

Example

Versioning = Enabled

↓

MFA Delete = Allowed

5. Who can enable MFA Delete?

Answer

Only the AWS account root user can enable or disable MFA Delete. IAM users and IAM roles cannot configure MFA Delete even if they have administrator permissions. This restriction provides additional security.

Example

Root User ✓

IAM Admin ✗

6. How does MFA Delete improve security?

Answer

MFA Delete requires something the user knows (password) and something the user has (OTP device). This two-factor authentication prevents unauthorized users from deleting important object versions. Even if AWS credentials are stolen, the attacker still needs the MFA device.

Example

Password + OTP
= Delete Allowed

7. Can MFA Delete protect against ransomware?

Answer

Yes. MFA Delete helps protect versioned backups from ransomware attacks. If an attacker gains AWS access, they still cannot permanently remove protected versions without the MFA device. Organizations often combine Versioning and MFA Delete for stronger protection.

Example

Attacker Deletes File
↓
Old Versions Still Protected

8. Does MFA Delete affect normal uploads?

Answer

No. MFA Delete only affects sensitive operations like deleting object versions or changing Versioning settings. Normal uploads, downloads, and object updates continue to work without MFA verification.

Example

Upload report.pdf
↓
No OTP Needed

9. What happens if MFA code is incorrect?

Answer

AWS rejects the operation. The object version remains unchanged and protected. The user must provide a valid MFA code to perform the requested action. This prevents unauthorized deletions.

Example

Wrong OTP
↓
Access Denied

10. What is the best practice for MFA Delete?

Answer

Enable Versioning and MFA Delete on buckets containing backups, financial data, legal documents, and critical business records. Store MFA devices securely and restrict root account usage. Combine MFA Delete with Lifecycle Policies and Replication for maximum protection.

Example

Versioning + MFA Delete + Replication
= Strong Data Protection

Amazon S3 Replication – Complete Explanation

What is S3 Replication?

Definition

Amazon S3 Replication is a feature that automatically copies objects from one S3 bucket to another S3 bucket. Whenever a new object is uploaded to the source bucket, AWS automatically creates a copy of that object in the destination bucket.

It is mainly used for:

- Backup
 - Disaster Recovery
 - High Availability
 - Compliance Requirements
 - Data Distribution
-

Example

Source Bucket

Mumbai Region
Bucket-A

Upload:

employee_data.pdf

AWS automatically copies it to:

Destination Bucket

Singapore Region
Bucket-B

Result:

employee_data.pdf

exists in both buckets.

Types of S3 Replication

1. Cross-Region Replication (CRR)

Definition

Cross-Region Replication copies objects between buckets located in different AWS Regions.

Example

Mumbai Region



Singapore Region

Use Cases

- Disaster Recovery
 - Global Applications
 - Regulatory Compliance
-

2. Same-Region Replication (SRR)

Definition

Same-Region Replication copies objects between buckets within the same AWS Region.

Example

Bucket-A (Mumbai)



Bucket-B (Mumbai)

Use Cases

- Log Aggregation
 - Internal Backups
 - Testing Environments
-

Prerequisites for Replication

1. Versioning Must Be Enabled

Replication requires Versioning to be enabled on:

Source Bucket

Versioning = Enabled

Destination Bucket

Versioning = Enabled

Without Versioning:

Replication = Not Possible

2. IAM Permissions

AWS needs permission to copy objects from the source bucket to the destination bucket.

How Replication Works

User Uploads File



Source Bucket



Replication Rule



Destination Bucket

AWS performs this process automatically.

What Gets Replicated?

The following items are copied:

- ✓ Objects
 - ✓ Object Metadata
 - ✓ Object Tags
 - ✓ ACLs (Access Control Lists)
 - ✓ Object Versions
-

What Does Not Replicate Automatically?

Existing Objects

Objects uploaded before enabling replication are not replicated by default.

Example

File uploaded yesterday

Replication enabled today

Result:

File will NOT replicate automatically

Only newly uploaded objects are replicated.

Replication Time Control (RTC)

Definition

Replication Time Control provides a service-level agreement that objects will be replicated within approximately 15 minutes.

Example

Upload Time: 10:00 AM

Object should be replicated before:

10:15 AM

Replication and Versioning

Example

Source Bucket:

report.pdf v1

report.pdf v2

report.pdf v3

Destination Bucket:

report.pdf v1

report.pdf v2

report.pdf v3

All versions can be replicated.

Replication and Delete Markers

When an object is deleted from a versioned bucket:

Delete Marker Created

Depending on the replication configuration, the delete marker can also be copied to the destination bucket.

Amazon S3 Storage Classes – Complete Notes

What is an S3 Storage Class?

Definition

Amazon S3 Storage Classes are different storage tiers designed for different access patterns, performance needs, availability requirements, and costs. Instead of storing all data at the same price, AWS provides multiple storage classes so that organizations can optimize costs based on how frequently data is accessed. Frequently accessed data can be stored in high-performance classes, while rarely accessed data can be moved to cheaper archival classes. Storage classes help reduce storage costs while maintaining durability and security. Lifecycle policies can automatically move

objects between storage classes. All storage classes provide high durability (11 9's). Choosing the correct storage class is an important part of AWS cost optimization.

Example

Frequently Used Website Images → S3 Standard

Monthly Backup Files → S3 Standard-IA

10-Year Archive Data → S3 Glacier Deep Archive

1. S3 Standard

Definition

S3 Standard is the default storage class and is designed for frequently accessed data. It provides low latency, high throughput, and high availability. Data is stored across multiple Availability Zones (AZs), making it highly durable and resilient. It is suitable for applications that require frequent read and write operations. S3 Standard is commonly used for websites, mobile applications, content distribution, analytics, and data lakes. It offers immediate access to data whenever needed. Although it is the most expensive storage class, it provides the highest performance. Organizations use it when data is accessed regularly.

Features

- Frequent access
- Low latency
- High throughput
- Multi-AZ storage
- Immediate retrieval
- 99.99% Availability
- 99.999999999% Durability

Example

Website Images

Application Data

User Uploaded Files

2. S3 Standard-Infrequent Access (Standard-IA)

Definition

S3 Standard-IA is designed for data that is accessed less frequently but still requires immediate access when needed. It provides the same durability as S3 Standard but at a lower storage cost. Since storage is cheaper, AWS charges a retrieval fee whenever data is accessed. Data is stored across multiple Availability Zones for high availability. This storage class is ideal for backups, disaster

recovery files, and long-term storage that occasionally needs access. It balances cost savings and fast retrieval.

Features

- Lower storage cost
- Multi-AZ storage
- Millisecond retrieval
- Retrieval charges apply
- High durability
- Suitable for backups

Example

Monthly Backups

Old Reports

Disaster Recovery Files

3. S3 One Zone-Infrequent Access (One Zone-IA)

Definition

S3 One Zone-IA is similar to Standard-IA but stores data in only one Availability Zone instead of multiple AZs. Because it uses a single AZ, it costs less than Standard-IA. It is suitable for data that can be recreated if lost. Organizations often use it for secondary backups and temporary files. Since it lacks multi-AZ redundancy, it is not recommended for critical production data. It still provides immediate access to stored objects.

Features

- Single AZ storage
- Lower cost
- Fast retrieval
- Retrieval fee applies
- Less resilient than Standard-IA

Example

Development Backups

Temporary Data

Re-creatable Files

4. S3 Intelligent-Tiering

Definition

S3 Intelligent-Tiering is designed for data with unpredictable access patterns. AWS automatically moves objects between different access tiers based on usage. Frequently accessed data stays in higher-performance tiers, while less-accessed data moves to lower-cost tiers. This helps optimize storage costs automatically. Users do not need to manually analyze access patterns. Intelligent-Tiering is ideal for applications where future access frequency is unknown.

Features

- Automatic tier movement
- Cost optimization
- No performance impact
- No operational overhead
- Multiple access tiers

Example

User Documents

Shared Files

Unknown Access Data

5. S3 Glacier Instant Retrieval

Definition

S3 Glacier Instant Retrieval is designed for archive data that is rarely accessed but still requires immediate retrieval. It provides lower storage costs than Standard-IA while maintaining millisecond access times. It is suitable for long-term archives that occasionally need fast access. Organizations use it for medical records, media archives, and compliance documents.

Features

- Archive storage
- Instant retrieval
- Lower cost
- High durability
- Millisecond access

Example

Medical Records

Historical Images

Archived Media Files

6. S3 Glacier Flexible Retrieval

Definition

S3 Glacier Flexible Retrieval is a low-cost archival storage class for data that is rarely accessed. Retrieval times vary from minutes to several hours depending on the selected retrieval option. It is suitable for archive data, compliance storage, and long-term backups. This storage class offers significant cost savings compared to Standard and IA classes. It is commonly used for regulatory archives and old business records.

Features

- Very low cost
- Archive storage
- Retrieval in minutes or hours
- Long-term retention
- High durability

Example

Old Backups
Audit Logs
Compliance Records

7. S3 Glacier Deep Archive

Definition

S3 Glacier Deep Archive is the lowest-cost storage class offered by Amazon S3. It is intended for data that is rarely, if ever, accessed but must be retained for many years. Retrieval can take several hours. It is commonly used for legal records, tax documents, healthcare archives, and long-term compliance data. It provides the same high durability as other S3 storage classes while minimizing storage costs.

Features

- Lowest storage cost
- Long-term retention
- Retrieval in hours
- Suitable for archives
- High durability

Example

Tax Records
Legal Documents
10-Year Archives

Storage Class Comparison

Storage Class	Access Pattern	Retrieval Time
Standard	Frequent	Milliseconds
Standard-IA	Infrequent	Milliseconds
One Zone-IA	Infrequent	Milliseconds
Intelligent-Tiering	Variable	Milliseconds
Glacier Instant Retrieval	Rare	Milliseconds
Glacier Flexible Retrieval	Rare	Minutes–Hours
Glacier Deep Archive	Very Rare	Hours

Important Interview Questions

Which S3 Storage Class is used for frequently accessed data?

S3 Standard

Which S3 Storage Class is the cheapest?

S3 Glacier Deep Archive

Which Storage Class automatically moves objects between tiers?

S3 Intelligent-Tiering

Which Storage Class stores data in only one Availability Zone?

S3 One Zone-IA

Which Storage Class provides instant archive retrieval?

S3 Glacier Instant Retrieval

Which Storage Class is best for long-term backups?

S3 Glacier Flexible Retrieval or Glacier Deep Archive

Quick Revision

S3 Standard → Frequently Accessed Data

S3 Standard-IA → Infrequently Accessed Data

S3 One Zone-IA → Single AZ Storage

S3 Intelligent-Tiering → Automatic Cost Optimization

S3 Glacier Instant Retrieval → Archive + Instant Access

S3 Glacier Flexible Retrieval → Archive + Minutes/Hours Retrieval

S3 Glacier Deep Archive → Lowest Cost Long-Term Archive